

ATARS v3.1

9 Verification Modules Working Together in One Pipeline

The contribution is the integration — not any one module individually

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github.com/Priyanshu-ux712/ATARS · MIT License · v3.1.0 (2026)

v2.0.0

Foundation

9

Verifiers

1

Pipeline

MNS

Trust Score

The Problem — Why One Verifier Is Not Enough

An LLM can fail in 9 distinct ways. Each is invisible to a verifier designed for a different one.

Wrong number

"83.4 instead of 91.2 $\mu\text{g}/\text{m}^3$ "

Blind spot: TCV, CCD, NSS miss it

Wrong month

"'peaked in July' (actual: Jan)"

Blind spot: RGV, CVCC miss it

False causation

"'humidity causes PM10 to rise'"

Blind spot: RGV, TCV miss it

Vague language

"'air quality was poor'"

Blind spot: RGV — no numbers to check

No uncertainty

"91.2 without a CI range"

Blind spot: All others miss it

Chemical error

"'NOx < NO2' (impossible)"

Blind spot: RGV, TCV, CCD miss it

No benchmark

"91.2 with no WHO comparison"

Blind spot: All text verifiers miss

Quality drift

"G_rate 92%→71% across runs"

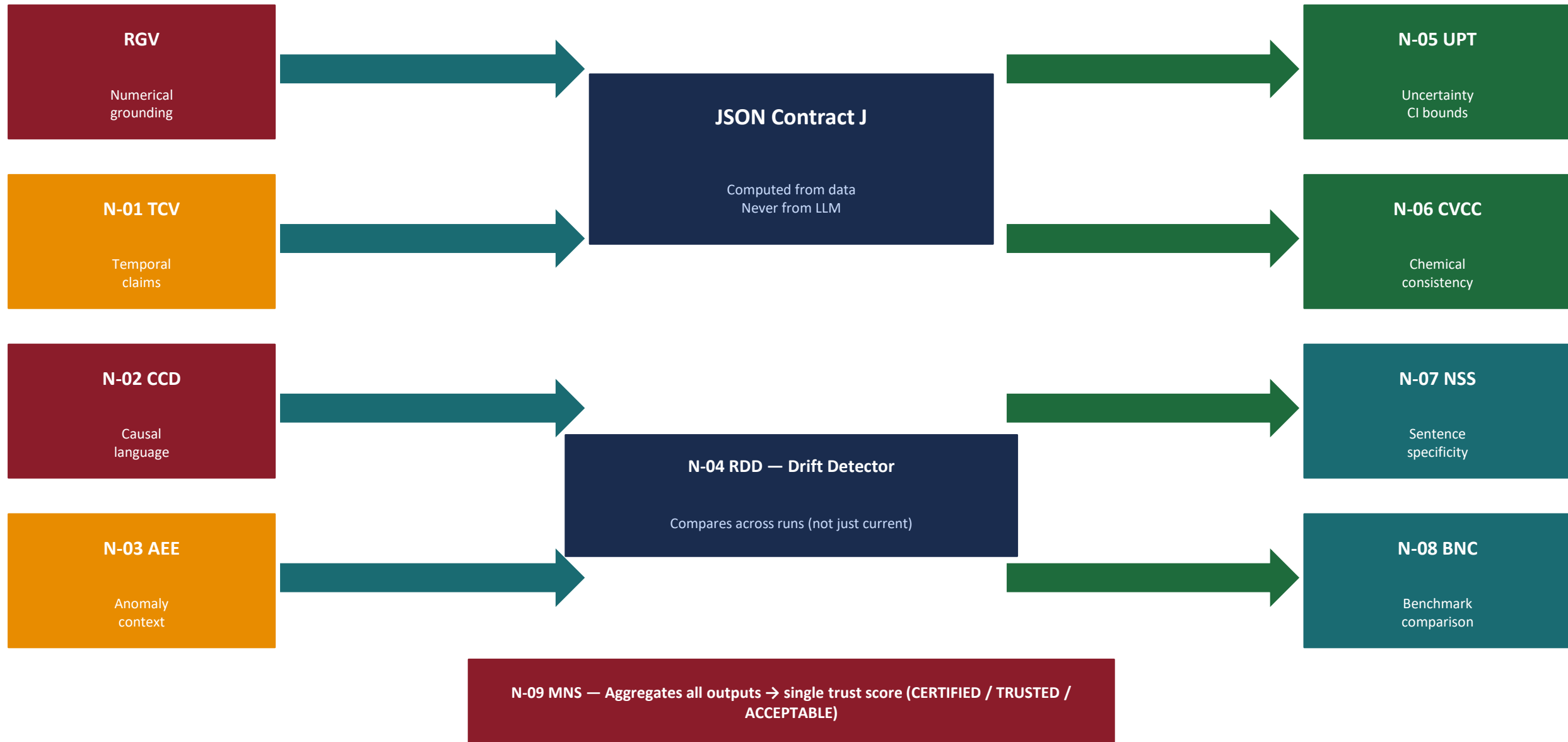
Blind spot: All per-run modules miss

No event context

"Anomaly unexplained"

Blind spot: All text verifiers miss

The System — All 9 Modules Share One Input: JSON Contract J



How They Complement Each Other — A Real Example

"High humidity in July caused PM10 to peak at 83.4 $\mu\text{g}/\text{m}^3$, significantly above normal levels."

This one sentence has 4 errors — caught by 4 different modules:

"83.4 $\mu\text{g}/\text{m}^3$ "

Wrong number
(J says 91.2)

RGV

91.2 vs 83.4 = 8.6% diff
> $\pm 3\%$ tolerance → FAIL

CAUGHT ✓

"in July"

Wrong month
(actual: January)

N-01 TCV

Claimed July
Actual peak: January → FAIL

CAUGHT ✓

"caused"

Causal language
(only r in J)

N-02 CCD

'caused' trigger found
No causal data in J → FLAGGED

CAUGHT ✓

"significantly above normal"

Vague — no number,
no benchmark

**N-07 NSS
+ N-08 BNC**

Specificity = 0.0
No WHO comparison → FLAGGED

CAUGHT ✓

No single module catches all 4. RGV misses the month. TCV misses the wrong number. CCD misses the number. NSS misses the causation. Together: complete coverage.

What Each Module Catches — and What It Misses

RGV	Wrong numbers in text	<i>Misses: Wrong month, causation, vagueness</i>
N-01 TCV	Wrong month/season claims	<i>Misses: Wrong numbers, causation</i>
N-02 CCD	Causal language (causes, due to)	<i>Misses: Wrong numbers, wrong months</i>
N-03 AEE	Unexplained anomaly events	<i>Misses: Text errors (it checks data, not text)</i>
N-04 RDD	Quality degradation across runs	<i>Misses: Errors within a single run</i>
N-05 UPT	Missing uncertainty bounds	<i>Misses: Wrong values, wrong time, causation</i>
N-06 CVCC	Chemical relationship errors (NO _x <NO ₂)	<i>Misses: Text-level errors</i>
N-07 NSS	Vague sentences without numbers	<i>Misses: Numerically correct but wrong values</i>
N-08 BNC	Missing benchmark context	<i>Misses: Text-level and temporal errors</i>
N-09 MNS	Nothing — it aggregates all outputs	<i>Misses: Everything (it relies on others)</i>

How ATARS Differs from FActScore and SelfCheckGPT

FActScore (Min et al. 2023)	SelfCheckGPT (Manakul et al. 2023)	ATARS v3.1 (This Work)
Checks text against Wikipedia / internet	Checks consistency across multiple LLM runs	Checks against computed JSON contract J
Requires retrieval from external sources	Requires multiple LLM API calls	Works offline no external API needed
Treats numbers like any other text token	Treats numbers like any other text token	Type-adaptive tolerance (conc. ≠ count ≠ score)
No domain-specific tolerance (5% vs 0.001%)	No domain-specific tolerance	9 modules covering 9 distinct failure modes
Single run only	Single run scope	N-04 RDD tracks quality across runs
No causal, temporal, or chemical checks	No structured data ground truth	Temporal, causal, chemical, specificity checks

Key distinction: ATARS uses a computed domain-specific ground truth (J), not internet retrieval. Numbers are verified with domain-aware tolerance, not as plain text.

N-09 MNS — Why a Single Trust Score Matters

Without MNS: a reader gets 9 separate scores and must combine them manually.

With MNS: one number answers 'How much should I trust this report?'

$$\text{MNS} = G_v2 \times 0.25 + G_v3 \times 0.20 + \text{TCV} \times 0.15 + \text{CCD} \times 0.15 + \text{CVCC} \times 0.10 + \text{Conf} \times 0.10 + \text{Sem} \times 0.05$$

CERTIFIED

≥ 0.90

All numerical claims verified, no causal errors,
no temporal errors, consistent across runs

TRUSTED

≥ 0.80

Minor issues only — suitable for policy use
with standard caveats

ACCEPTABLE

≥ 0.70

Some ungrounded claims — review before
use
in formal reports

REVIEW

≥ 0.55

Multiple failure modes triggered —
manual review required

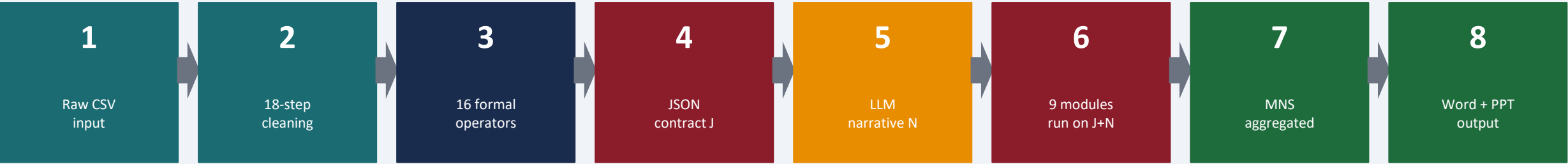
UNRELIABLE

< 0.55

Significant hallucination risk —
do not use without expert review

V2 G_rate=100% and V3=94.4% with --no-llm is expected: placeholder text verified against its own source is circular. Real scores only with Ollama active.

Pipeline Flow — Where Each Module Runs



Where each module fires:

After step 2	N-03 AEE runs on daily_df (not LLM text)
After step 4	N-06 CVCC checks J for chemical constraints
After step 5	RGV + N-01 TCV + N-02 CCD + N-07 NSS run on narrative N
After step 5	N-05 UPT annotates claims in N with CI from J
After step 5	N-08 BNC compares J values to WHO/NAAQS benchmarks
Across runs	N-04 RDD compares current audit log to previous logs
Step 7	N-09 MNS collects one output from each and computes trust score

Other New Features in v3.1 (Supporting the Core System)

18-Step Data Cleaning

Professional cleaning before any analysis. Flatline detection, spike filter, 3×IQR winsorization, 5-strategy imputation, chemical constraints, 7-dimensional quality score. Every step logged in Section 00b.

Statistical Inference Layer

Mann-Kendall trend test, normality tests (Shapiro-Wilk + D'Agostino), OLS with SE/t-stats/VIF/Durbin-Watson, seasonal decomposition, exceedance frequency analysis. Pure numpy — no statsmodels.

Open-Meteo Weather (18 vars)

Free ERA5 reanalysis + forecast. No API key. Batch-fetches all dates. SSL disabled for Windows compatibility. 18 derived indices: inversion, fog, dust, smog, stagnant, pollution risk 0–6.

Automated 19-Slide PPT

`generate_summary_ppt()` runs with every pipeline call. Slide 4: two-column quality layout. Slide 15: 4 live KPI cards (Q(D) / G_rate / MNS). No manual slide work required.

The Contribution in One Sentence:

"The first automated environmental reporting system to integrate 9 complementary verification modules into a single pipeline, where each module catches what the others cannot."

9 failure modes → 9 modules → 1 pipeline → 1 trust score

9

Modules

18

Cleaning Steps

MNS

Trust Metric

v2+v3

Both Versions

Code: github.com/Priyanshu-ux712/ATARS

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References: FActScore (Min et al. 2023) · SelfCheckGPT (Manakul et al. 2023) · WHO AQG 2021 · Open-Meteo ERA5 · Liu et al. 2008 (Isolation Forest)